Department of Electrical and Computer Engineering
Queen's University

ELEC-271 Digital Systems

Problem Set 3

1. Find the minimum-cost SOP and POS forms for the function $f(x_1, x_2, x_3) = \sum m(1,2,3,5)$.

2. Repeat question one for the function $f(x_1, x_2, x_3, x_4) = \sum m(0,2,8,9,10,15) + D(1,3,6,7)$.

3. Repeat question one for the function $f(x_1,...,x_5) = \prod M(1,4,6,7,9,12,15,17,20,21,22,23,28,31)$.

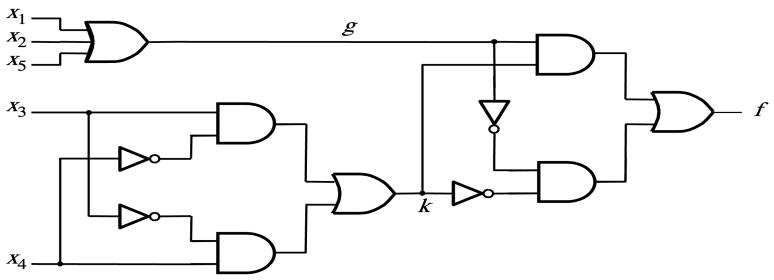
4. Derive a minimum-cost SOP and POS realization of the four-variable function that is equal to 1 if exactly two or three of its variables are equal to 1; otherwise, it is equal to 0.

5. A circuit with two outputs has to implement the following functions

$$f(x_1,...,x_4) = \sum m(0,2,4,6,7,9) + D(10,11)$$
$$g(x_1,...,x_4) = \sum m(2,4,9,10,15) + D(0,13,14)$$

   Design the minimum-cost circuit and compare its cost with combined costs of two circuits that implement f and g separately. Assume that the input variables are available in both uncomplemented and complemented forms.

6. Implement the following logic circuit using NAND gates only.



7. Find the simplest realization of the function $f(x_1, x_2, x_3, x_4) = \sum m(0,3,4,7,9,10,13,14)$, assuming that the logic gates have a maximum fan-in of two.

8. Use functional decomposition to find the best implementation of the function $f(x_1,...,x_5) = \sum m(1,2,7,9,10,18,19,25,31) + D(0,15,20,26)$. How does your implementation compare with the lowest-cost SOP implementation? Give the cost.

9. Consider the logic expressions

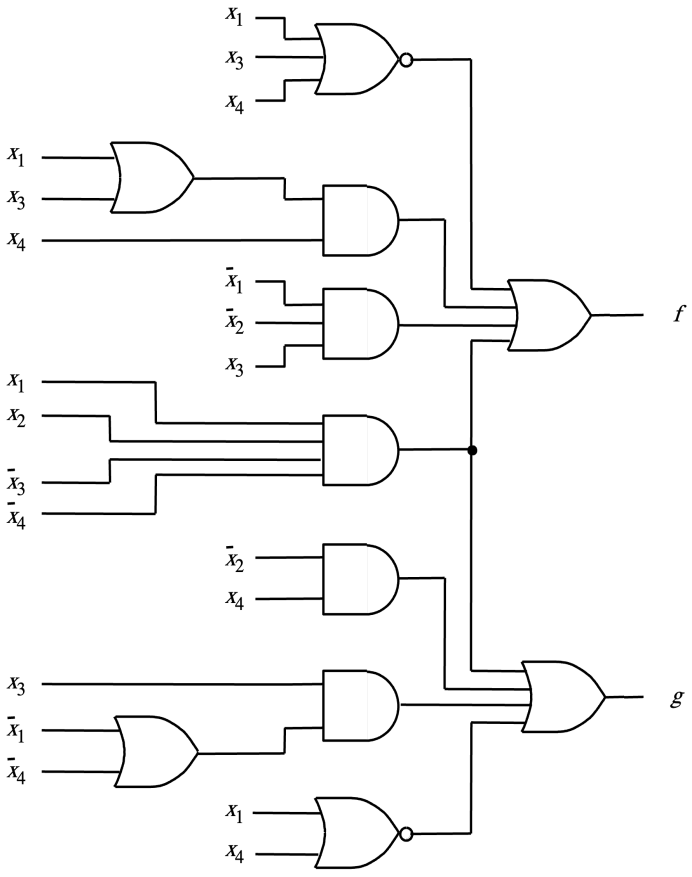
$$f = x_1\bar{x}_2\bar{x}_3 + x_2x_4 + x_1\bar{x}_2\bar{x}_4 + \bar{x}_2\bar{x}_3\bar{x}_4 + \bar{x}_1x_2x_3$$

$$g = (\bar{x}_2 + x_3 + x_4)(\bar{x}_1 + \bar{x}_2 + x_4)(x_2 + \bar{x}_3 + \bar{x}_4)(x_1 + x_2 + \bar{x}_3)(x_1 + x_2 + \bar{x}_4)$$

Prove or disprove that $f = g$.

10. Find the minimum-cost circuit consisting of only two-input NAND gates for the function $f(x_1, x_2, x_3, x_4) = \sum m(0,1,2,3,4,6,8,9,12)$. Assume that the input variables are available in both uncomplemented and complemented forms.
Hint: Consider the complement of the function.

11. Consider the following circuit, which implements function f and g. What is the cost of this circuit, assuming that the input variables are available in both true and complemented forms? Redeisgn the circuit to implement the same functions, but at as low a cost as possible. What is the cost of your circuit?



12. Write VHDL code to implement the circuit in figure of question 6.